Ecorobotix Data Sample

There are two types of data that the DigAg department manipulates: raw and processed data. The raw data is exactly what is sent by the machine to the cloud. The Processed data is everything that we extract from it.

Raw Data from the Machine

mission_data.json

- "start_time_ms": 1746773216805 -> Float,
- "end_time_ms": 1746784879000 -> Float,
- "robot_id": "erxbot-XXXX" -> String,
- "field_owner": "xxxxx" -> String, entered manually by the user,
- "field_name": "xxxxx" -> String, entered manually by the user,
- "travelled_distance_m": 14733.165 -> Float in meters,
- "covered_area_m2": 83945.99 -> Float in square meters,
- "sprayed_fluid_l": 41.79406 -> Float in liters,
- "context_eppo": "BEAVA" -> String,
- "target_eppo": "WEED" -> String,
- "is_selective": false -> Bool,
- "target_fluid_surface_density_lpha": 250.0 -> Float in Liter per hectare,
- "setpoint_pressure_bar": 2.7 -> Float,
- "cultivation_method": "FLAT" -> String,
- "inference_model": "BEAVA/9" -> String

mission_time_series.csv

- timestamp_ms -> Float
- system_state -> String, SPRAYING or IDLE
- gps_latitude_rad -> Float

- gps_longitude_rad -> Float
- robot_pose_x_m -> Float
- robot_pose_y_m -> Float
- robot_pose_theta_rad -> Float
- robot_speed_x_mps -> Float
- robot_speed_y_mps -> Float
- robot_speed_omega_radps -> Float
- unit0_enabled -> Bool
- unit1_enabled -> Bool
- unit2_enabled -> Bool
- unit0_fluid_quantity_l -> Float
- unit1_fluid_quantity_l -> Float
- unit2_fluid_quantity_l -> Float
- target_coverage_pcent -> Float
- avoid_coverage_pcent -> Float

Images

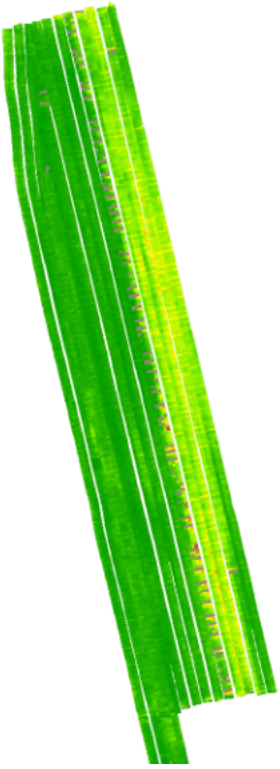
Sent by robot (sampling rate to define) -> .png

Transformed Data

Heatmap

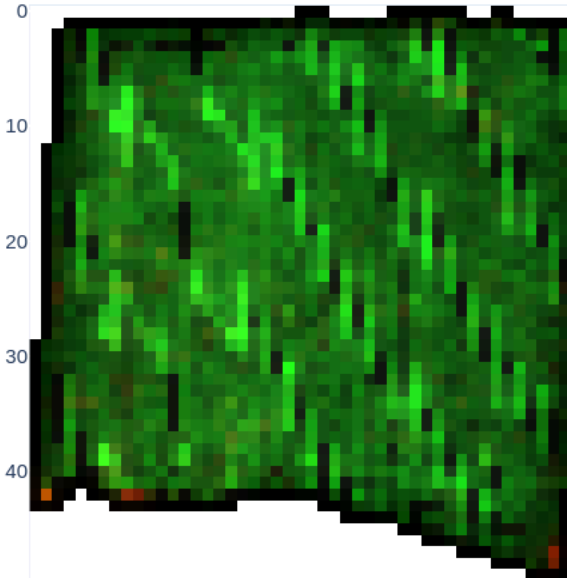
Processed and Unprocessed (color scaling)

- Liquid application .png



- Avoid (crop) and target (weed) density heatmap .png

allce(green) - weed(red)



mission_metadata.json

- "num_gps_0_0_removed": 0 -> Int number of missing GPS data,
- "no_valid_GPS_data": false -> Bool if no valid GPS data in entire mission,

- "max_lat": 52.70565 -> Float,
- "min_lat": 52.702878 -> Float,
- "max_long": 5.7953279 -> Float,
- "min_long": 5.7906080 -> Float,
- "area_analyzed": 7.896 -> Float in m2,
- "average_density": 151.174 -> Float in Liter per Hectar
- "crop_weed_correlation": 0.62 -> Float between 0-1
- "weed_liquid_correlation": 0.74 -> Float between 0-1